

1. Explain how negative feedback works to maintain homeostasis

Negative feedback is a mechanism used by the body to maintain homeostasis, ensuring that internal conditions remain within a stable range despite external changes. It works by detecting deviations from a set point and triggering responses that counteract the change, returning the system to its normal state. Through the process of: Stimulus - sensor (Receptor) - control center - effector - response - feedback.

2. Define depolarization and hyperpolarization. For a neuron, if the permeability of a membrane to sodium increases, explain the change in the membrane potential

① Depolarization: refers to a decrease in the difference in electrical charge across a cell membrane. This occurs when the inside of the neuron becomes less negative (closer to 0 or even positive) compared to the outside. Typically, this happens due to an influx of positively charged ions like sodium (Na^+) into the cell.

② Hyperpolarization: refers to an increase in the difference in electrical charge across a cell membrane, making the inside of the neuron more negative compared to the outside. This usually results from the efflux of positively charged ions like potassium (K^+) or the influx of negatively charged ions like chloride (Cl^-).

③ If the sodium permeability increases, Na^+ are more concentrated outside the cell compared to inside, and the resting membrane potential is negative (around -70mV). When the permeability increases, sodium ions flow into the cell following their electrochemical gradient.

As the sodium ions enter the cell, the inside of the neuron becomes less negative. This leads to depolarization of the membrane potential.

If it reaches the threshold, it will trigger the Action potential.

3. What is the difference between a carrier and a channel.

① Carrier proteins

Mechanism : Carrier proteins undergo conformational changes to transport specific molecules across the membrane. They bind to the molecule, change shape, and release the molecule on the other side of the membrane.

Specificity : Usually transporting only one or a few related molecules or ions.

Speed : Slower compared to channels because of the conformation change required for transport.

Transport : Requires energy, usually from ATP or electrochemical gradient.

② Channel proteins

Mechanism : Channel proteins form a hydrophilic pore that allows specific molecules or ions to move across the membrane without binding to the channel.

Specificity : Less specific than carriers but often selective for certain ions or molecules based on size and charge.

Speed : Faster compared to carriers since molecules move directly through the pore.

Transport Type : Mediate passive transport only.

r. Explain the ionic basis of an action potential

An action potential begins with sodium ion influx during depolarisation, followed by potassium ion efflux during repolarization. Resting potential is restored as ion concentrations return to baseline through active transport.

5. Name the three different gating mechanisms for ion channels.

1. Voltage-gated
2. Ligand-gated
3. Mechanically-gated.

6. Understand the differences between chemical and electrical synapses.

Chemical synapses : Use neurotransmitters to transmit signals between neurons, allowing for flexibility and signal amplification.

Electrical synapses : Direct transmission of ions and small molecules via gap junctions, providing faster signal propagation.

7. Describe the sequence of events that occurs in the crossbridge cycle.

1. ATP binds to the myosin head, detaching it from actin.
2. ATP is hydrolyzed, ~~energizing~~ energizing the myosin head.
3. Myosin binds to actin, forming a crossbridge.
4. power stroke occurs as myosin pulls actin, releasing ADP.
5. ATP binds again to detach myosin and restart the cycle.

8. What is Starling's law

Starling's law states that the stroke volume of the heart increases with an increase in the volume of blood filling the heart (preload), due to the stretch of the myocardial fibers

9. Trace the path of action potentials through the conduction system of the heart.

SA node → Atrial muscle → AV node → Atrioventricular bundle
→ Bundle branches → Purkinje fibers → Ventricular muscle.

10. Describe the ionic mechanism of action potential in Ventricular myocardial cells

- ① Rapid sodium influx causes depolarization.
- ② plateau phase occurs due to slow calcium influx and delayed potassium efflux
- ③ Repolarization is completed by potassium efflux.

11. Describe the components of the electrocardiogram (ECG)

P wave : Atrial depolarization.

QRS complex : Ventricular depolarization

T wave : Ventricular repolarization,

12. Describe what the arterial baroreceptor reflex is.

It is a feedback mechanism where baroreceptors in the carotid sinus and aortic arch detect changes in blood pressure and adjust heart rate and vascular resistance to maintain stable blood pressure.

13. Arrange the order of electrical impulses through the heart.

SA - node - Atrial muscle - AV node - Atrioventricular bundle - Bundle branches - Purkinje fibers - Ventricular muscle.

14. Which stimuli activate peripheral and central chemoreceptors.

P: Low $P-O_2$, high $P-CO_2$, low P^H

C: High $P-CO_2$, low pH in cerebrospinal fluid.

15. Sources and function of surfactant.

Sources: produced by type II alveolar cells in the lungs.

Function: Reduces surface tension in alveoli, preventing collapse during exhalation.

16. List the different lung volumes and capacities.

Volumes: Tidal volume, inspiratory reserve volume, expiratory reserve volume, residual volume.

Capacities: Vital capacity, inspiratory capacity, functional residual capacity, total lung capacity.

17. Describe the processes of respiration

Aerobic respiration. : Glycolysis - Link Reaction - Krebs cycle
- Electron transport chain - total ATP yield

18. How changes in $P-O_2$ and $P-CO_2$ in lung tissues alter ventilation and mechanism.

Low $P-O_2$ primarily activates peripheral chemoreceptors, while high $P-CO_2$ strongly stimulates central chemoreceptors. Both responses are integrated in the medulla, driving respiratory adjustment. This feedback mechanism ensures adequate oxygen delivery and carbon dioxide removal, maintaining arterial blood gas homeostasis.

19. Describe glucose absorption in the small intestine

Glucose is absorbed through sodium-glucose symporters (SGLT) into epithelial cells and exit into the bloodstream via GLUT2 transporters.

20. process of urine production

Filtration in the glomerulus

Reabsorption in tubules

Secretion of waste

Excretion through ureters to bladders.